

C - Basics

```
#include <stdio.h>
int main()
{
    // Hello World

    /*Multiple
    Line
    comment*/

    printf("Hello, World!\n\n");
    printf("Hello %d %d\n", 20, 21); //int
    printf("Hello %.2f\n", 93.7); //float
    printf("a = %d, b = %d, sum = a + b = %d + %d = %d \n", 3,
5, 3, 5, 3+5);
    printf("%d %% %d = %d \n", 5, 2, 5%2);
    printf("%s", "Help"); //string
    return 0;
}
```

```
#include <stdio.h>
int main()
{
    int age;
    age = 30;
    double temp = 26.5;

    double height, width;
    scanf("%lf", &height);
    scanf("%lf", &width);
    printf("Area = %.2lf", (height*width));

    //casting
}
```

```

    int num1 = 5;
    int num2 = 2;
    double result;
    result = (double)num1/num2;
    printf("%.2lf", result);

    (int)5.11

    return 0;
}

```

Celsius to Fahrenheit

```

#include<stdio.h>
int main()
{
    double ftemp, ctemp;
    printf("Enter degrees in Celcius: ");
    scanf("%lf", &ctemp);
    ftemp = ctemp * 1.8 + 32;
    printf("Conversion of %.2lf Celcius is %.2lf Fahrenheit
\n", ctemp, ftemp);
    return 0;
}

```

Strings

```

#include <stdio.h>
#include <stdlib.h>
int main()
{
    char name[10];
    int age;
    printf("Enter your first name and age: \n");
    scanf("%s %d", name, &age);
    printf("You entered: %s %d",name,age);
}

```

```
    return 0;
}
```

How much do I earn hours*per hour

```
#include <stdio.h>
#include <stdlib.h>
int main()

{
    double salary;
    int hours;
    printf("How many hours did you work? ");
    scanf("%d", &hours);
    printf("What is your hourly rate? ");
    scanf("%lf", &salary);
    printf("You earned %.2lf for working %d", (double)
(salary*hours),hours);
    return 0;
}
```

Time taken in hours and mins depending on speed

```
#include <stdio.h>
#include <stdlib.h>
#include <math.h>

int main()
{
    int distance, speed;
    printf("Enter the distance you are travelling in miles:
");
    scanf("%d", &distance);
    printf("Enter the speed your travelling in miles per
hour: ");
    scanf("%d", &speed);
```

```

    float speedInMinutes = speed / 60.0;
    float timeInMinutesRemaining = fmod(distance, speed) /
speedInMinutes;
    printf("It will take %d hours and %.2f minutes",
(distance/speed), timeInMinutesRemaining);
    return 0;
}

```

Seconds to hours/mins/secs

```

#include <stdio.h>
#include <stdlib.h>
#include <math.h>

int main()
{
    printf("Enter seconds: ");
    int seconds, hours;
    scanf("%d", &seconds);
    printf("Total hours = %d\nTotal minutes = %d\nRemaining
seconds = %d", seconds/3600, (seconds%3600)/60,
(seconds%3600)%60);
    return 0;
}

```